

Licensed Electrician's Theory (LET) Assessment Sample Paper Marking Guide 2024

AS/NZS 3000 Wiring Rules

Question 1

75mm (2 marks)

Clause 4.4.2.2 (b) (2 marks)

Question 2

No (2 marks)

Clause 5.4.5 Exceptions 1 (2 marks)

Question 3

Persons or livestock (2 marks)

Clause 2.7.3 (c) (2 marks)

Question 4

No, lifts shall be supplied by a dedicated circuit (2 marks)

Clause 4.20.2 (2 marks) Correct for non-Safety Services only.

AS/NZS 3012 Construction and Demolition Sites

Question 5

A qualified person (2 marks)

Clause 3.3.1 (2 marks)

Question 6

Electrical regulators (2 marks)

Clause 3.9 (2 marks)

Electrical Safety (General) Regulations 2019

Question 7

A written report of the incident (2 marks)

Regulation 401(3) (2 marks)

Electrical Shock Survival

Question 8

Check safety to yourself and others (Check for danger) (2 marks)

30 compressions and 2 breaths (2 marks)

Cable Selection

Question 9

Part (i)

Table 3(4) Item 4 (1 Mark - item number is optional do not deduct marks)

Table 14 Col 25 (2 Marks)

Table 26(2) Col 3 Derating for Touching 2 circuits = 0.9 (1 Mark)

Table 28(2) Col 3 Derating for Depth 1m = 0.96 (1 Mark)

2 cables in parallel $260 / 2 = 130\text{A}$ per cable

$35\text{mm}^2 = 130\text{A}$

$130 \times 0.9 \times 0.96 = 112.32\text{A}$

$50\text{mm}^2 = 155\text{A}$

$155 \times 0.9 \times 0.96 = 133.92$

Answer 50mm^2

Part (ii)

Table 28(2) Col 3 Derating for Depth 0.6m = 0.99

$155 \times 0.9 \times 0.99 = 138.11\text{A}$

Answer = 50mm^2 (1 Mark for all)

(Deduct 1 Mark for no or incorrect units)

DC Circuits

Question 10

(i) Meter X – 6.50W

(ii) Meter Y – 0.38A

(iii) Meter Z – 24.32V

Deduct 1 mark for no or incorrect units.

Maximum Demand**Question 11**

Table C2 Column 2 (1 mark)

- 1 - 5kW Electric roller door
- 1 - 12A reverse cycle air conditioner
- 56 - 15W LED lighting points
- 1 - 5.8kW cooking appliance
- 26 - 10A double socket outlets

Domestic Residence.

Table C2 Column 2

Equipment	Load Group	Calculation	Maximum Demand
1 – 5kW Electric roller door	(d)	Full connected load 21.74A $5000/230 = 21.74A$	21.74A (1 mark)
1 - 12A reverse cycle air conditioner	(c)(i)	50% connected load $12A \times 50\% = 6A$	6A (1 mark)
56 - 15W LED lighting points	A	75% connected load $56 \times 15 = 840W$ $840/230 = 3.65A$ $3.65 \times 75\% = 2.74A$	2.74A (1 mark)
1 - 5.8kW cooking appliance	(c)(i)	Full connected load $5800/230 = 25.22A$	25.22A (2 mark)
26 - 10A double socket outlets 52 Points total	(b)(ii)	$1000W + (100 \times 51) = 6100$ $6100/230 = 26.52A$	26.52A (1 mark)
Total Maximum Demand			82.22A (1 mark)

Deduct 1 mark for no or incorrect units on total, deduct 1 mark for no or incorrect load groups.

Voltage Drop**Question 12****Consumer's Mains**

Table 41 Column 10 (1 mark)

 $V_c 1.25$ (1 mark) $V_d = (11 \times 130 \times 1.25)/1000$ $V_d 1.79V$ (1 mark)**Sub-mains**

Table 41 Column 8 (1 mark)

 $V_c 2.55$ (1 mark) $V_d = (22 \times 48 \times 2.55)/1000$ $V_d 2.69V$ (1 mark)**Final Sub-circuit**

Table 42 Column 6 (1 mark)

 $V_c 6.49$ (1 mark) $V_d = (27 \times 29 \times 6.49)/1000$ $V_d 5.08V$ (1 mark)Total Voltage Drop = $1.79 + 2.69 + 5.08 = 9.56 V$ (1 mark)

Deduct 1 mark for no or incorrect units on total.

Deduct 1 mark for no or incorrect table number/s.

Overload and Short Circuit Calculations

Question 13

Overcurrent divided by MCB current rating = 2 (1 mark)

Minimum Time = Accept 26 - 28 seconds (1 mark)

Maximum Time = Accept 114 - 117 seconds (1 mark)

Deduct 1 mark for no or incorrect time unit.

Question 14

Transformer impedence

230/18000 (2 marks)

0.01278Ω (1 mark) Answer must be to **5 decimal places**.

Main switchboard prospective fault

230/ (0.01278 +0.0084) (2 marks)

10,859A (1 mark)

Distribution board prospective fault

230/ (0.01278 +0.0084+ 0.0038) (2 marks)

9,207A (1 mark)

Deduct 1 mark for no or incorrect units in final answer.

Residual Current Devices

Question 15

16A

Motor and Starters

Question 16

B (2 marks)

AS/NZS 4836:2023

Question 17

Through an RCD or an isolating transformer (2 marks)

Clause number: 8.3.2 (2 marks)

Installation Defects - Non Domestic

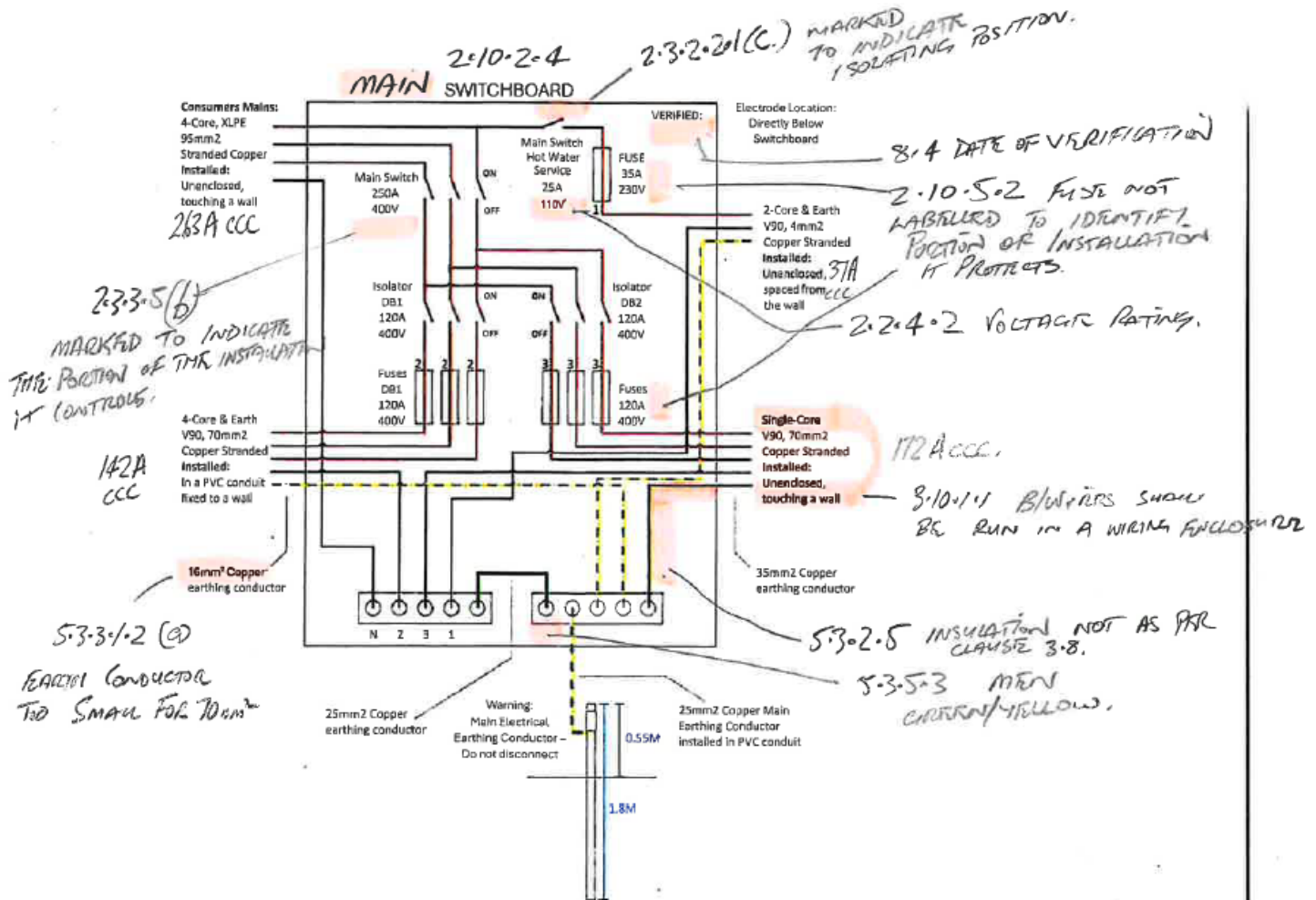
Question 18

2 marks for correct defect, 1 mark for the correct clause.

Only some of the defects have been listed below, there are more than 10 defects in the diagram. All correct defects and clauses will be awarded marks.

Only accept the first 5 defects a candidate has listed.

Installation Faults - Non Domestic



The drawing above shows the MAIN SWITCHBOARD for an Industrial Installation.
The drawing contains contraventions to the Wiring Rules

The MAIN SWITCHBOARD is supplied directly from a pole-mounted transformer via consumers mains installed unenclosed on a cable tray fixed directly to a wall and enters directly into the switchboard enclosure.

It supplies two (2) distribution boards, each having a connected load with a calculated maximum demand of 120A per phase, and a single-phase hot water heater rated at 25A. The water heater has been deemed not at increased risk of electric shock.

All other cables are installed as they are described on the drawing.

All fuses are HRC type.

All screws in bars and links are 75% of the tunnel terminal diameter.

A metal partition separates all switchgear from each other.

The Supply Authority has provided Short Circuit protection of the Point of Supply on the Consumers Mains.